

THE UNIVERSITY OF AZAD JAMMU & KASHMIR, MUZAFFARABAD

Department of Software Engineering

LAB 05

Windows GUI Controls: Login Page and Registration Form (WPF / C#)

Submitted to: Engr. Dr. Asma Javaid
Submitted By: Jawahir Ali
Session: 2024-28
Roll No: 2024-SE-34
Date: 27-10-2026

Objective

To design Windows GUI applications using common controls (TextBox, PasswordBox, RadioButton, DatePicker, CheckBox, Button), apply validation rules and handle the Click event to display a MessageBox.

Tools Used

- Visual Studio 2022
- C# with WPF (.NET) application
- Language: XAML for the design, C# for the code-behind

Question 1: User Login Page

Problem Statement

Create a GUI for a User Login page with one field for CNIC and one for Password. Use hard-coded validation.

Steps

- Create a new WPF App project named Lab05 and add a Window named LoginWindow.
- Place a TextBox (txtCnic), a PasswordBox (txtPassword) and a Button (btnLogin) with labels.
- Double-click the Login button to create the Click event.
- Compare the entered values with the hard-coded CNIC 12345-1234567-1 and password admin123.

XAML Code (LoginWindow.xaml)

```
<Window x:Class="Lab05.LoginWindow"
        xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
        xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
        Title="User Login" Height="300" Width="420"
        WindowStartupLocation="CenterScreen" ResizeMode="NoResize">
    <Grid>
        <TextBlock Text="USER LOGIN" FontSize="22" FontWeight="Bold"
                   HorizontalAlignment="Center" Margin="0,30,0,0"/>

        <TextBlock Text="CNIC:" FontSize="14" Margin="40,110,0,0"
                   HorizontalAlignment="Left" VerticalAlignment="Top"/>
        <TextBox x:Name="txtCnic" Width="240" Height="28" FontSize="14"
                 Margin="130,106,0,0" HorizontalAlignment="Left"
                 VerticalAlignment="Top" MaxLength="15"/>

        <TextBlock Text="Password:" FontSize="14" Margin="40,160,0,0"
                   HorizontalAlignment="Left" VerticalAlignment="Top"/>
        <PasswordBox x:Name="txtPassword" Width="240" Height="28" FontSize="14"
                     Margin="130,156,0,0" HorizontalAlignment="Left"
                     VerticalAlignment="Top"/>

        <Button x:Name="btnLogin" Content="Login" Width="100" Height="32"
                Margin="0,215,0,0" HorizontalAlignment="Center"
                VerticalAlignment="Top" Click="btnLogin_Click"/>
    </Grid>
</Window>
```

C# Code (LoginWindow.xaml.cs)

```
using System.Windows;

namespace Lab05
{
    public partial class LoginWindow : Window
    {
        // Hard-coded credentials
```

```
        private const string ValidCnic     = "12345-1234567-1";
        private const string ValidPassword = "admin123";

        public LoginWindow()
        {
            InitializeComponent();
        }

        private void btnLogin_Click(object sender, RoutedEventArgs e)
        {
            string cnic = txtCnic.Text.Trim();
            string pass = txtPassword.Password;

            // Empty field check
            if (cnic == "" || pass == "")
            {
                MessageBox.Show("Please enter CNIC and Password.",
                    "Login", MessageBoxButton.OK, MessageBoxImage.Warning);
                return;
            }

            // Hard-coded validation
            if (cnic == ValidCnic && pass == ValidPassword)
            {
                MessageBox.Show("Login Successful!\nWelcome, User.",
                    "Login", MessageBoxButton.OK, MessageBoxImage.Information);
            }
            else
            {
                MessageBox.Show("Invalid CNIC or Password!",
                    "Login Failed", MessageBoxButton.OK, MessageBoxImage.Error);
                txtPassword.Clear();
                txtCnic.Focus();
            }
        }
    }
}
```

Output

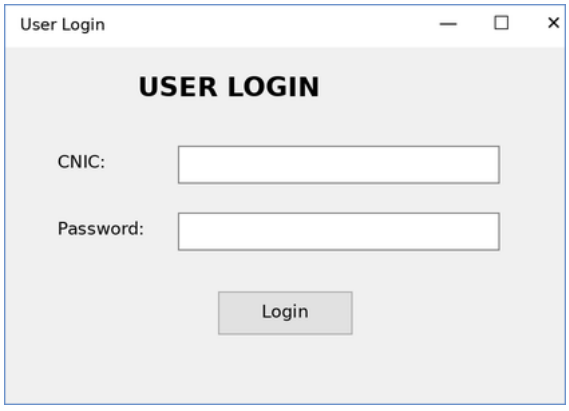


Figure 1.1: Login window on start-up

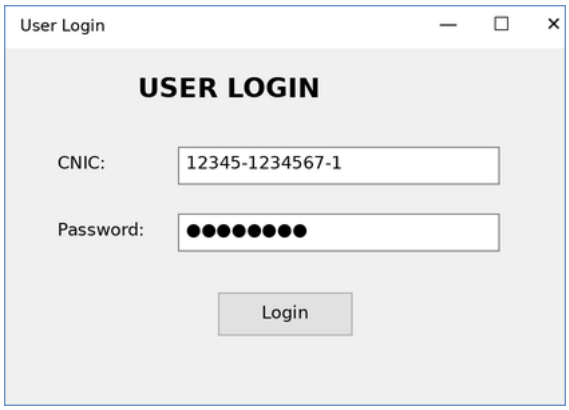


Figure 1.2: Valid CNIC and password entered

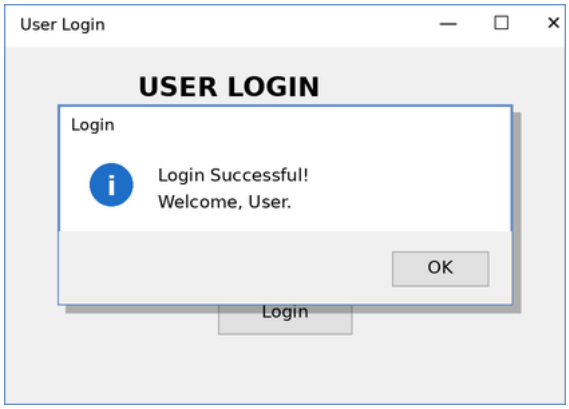


Figure 1.3: Successful login message

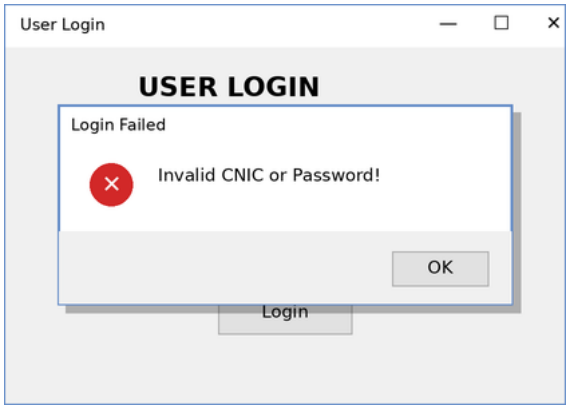


Figure 1.4: Invalid credentials error message

Question 2: Registration Form

Problem Statement

Create a GUI with a First Name and Last Name TextBox (mandatory), Gender RadioButtons, a DOB DatePicker, Hobbies CheckBoxes and a Submit button, following the given validation rules.

Controls Used

Control	Name	Purpose
TextBox	txtFirstName	First Name (mandatory *)
TextBox	txtLastName	Last Name (mandatory *)
RadioButton	rbMale / rbFemale	Gender (Male default)
DatePicker	dpDob	Date of Birth
CheckBox	chkCricket, chkTv, chkBooks, chkMusic, chkTravel	Hobbies
Button	btnSubmit	Validate and submit

Validation Rules Implemented

- First Name and Last Name must not be empty.
- Male radio button is checked by default.
- DOB default is today's date; DisplayDateStart = 1 Oct 1950 and DisplayDateEnd = today.
- On Submit, if valid, an Information MessageBox shows: welcome Firstname Lastname.

XAML Code (MainWindow.xaml)

```
<Window x:Class="Lab05.MainWindow"
        xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
        xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
        Title="Registration Form" Height="480" Width="460"
        WindowStartupLocation="CenterScreen" ResizeMode="NoResize">
    <Grid Margin="30">
        <Grid.RowDefinitions>
            <RowDefinition Height="45"/>   <!-- First name -->
            <RowDefinition Height="45"/>   <!-- Last name  -->
            <RowDefinition Height="45"/>   <!-- Gender     -->
            <RowDefinition Height="50"/>   <!-- DOB        -->
            <RowDefinition Height="170"/>  <!-- Hobbies    -->
            <RowDefinition Height="*"/>    <!-- Submit     -->
        </Grid.RowDefinitions>
        <Grid.ColumnDefinitions>
            <ColumnDefinition Width="120"/>
            <ColumnDefinition Width="*"/>
        </Grid.ColumnDefinitions>

        <TextBlock Grid.Row="0" Text="First Name *" VerticalAlignment="Center"/>
        <TextBox   Grid.Row="0" Grid.Column="1" x:Name="txtFirstName"
                   Height="28" VerticalContentAlignment="Center"/>

        <TextBlock Grid.Row="1" Text="Last Name *" VerticalAlignment="Center"/>
        <TextBox   Grid.Row="1" Grid.Column="1" x:Name="txtLastName"
                   Height="28" VerticalContentAlignment="Center"/>

        <TextBlock Grid.Row="2" Text="Gender" VerticalAlignment="Center"/>
        <StackPanel Grid.Row="2" Grid.Column="1" Orientation="Horizontal"
                    VerticalAlignment="Center">
            <RadioButton x:Name="rbMale"   Content="Male"   GroupName="Gender"
```

```
                        Margin="0,0,40,0" IsChecked="True"/>
            <RadioButton x:Name="rbFemale" Content="Female" GroupName="Gender"/>
        </StackPanel>

        <TextBlock Grid.Row="3" Text="Date of Birth" VerticalAlignment="Center"/>
        <DatePicker Grid.Row="3" Grid.Column="1" x:Name="dpDob"
                    Width="180" Height="28" HorizontalAlignment="Left"/>

        <TextBlock Grid.Row="4" Text="Hobbies" VerticalAlignment="Top"/>
        <StackPanel Grid.Row="4" Grid.Column="1">
            <CheckBox x:Name="chkCricket" Content="Playing Cricket"
Margin="0,0,0,12"/>
            <CheckBox x:Name="chkTv"      Content="Watching TV"
Margin="0,0,0,12"/>
            <CheckBox x:Name="chkBooks"   Content="Reading Books"
Margin="0,0,0,12"/>
            <CheckBox x:Name="chkMusic"   Content="Listening Music"
Margin="0,0,0,12"/>
            <CheckBox x:Name="chkTravel"  Content="Travelling to New Places"/>
        </StackPanel>

        <Button Grid.Row="5" Grid.ColumnSpan="2" x:Name="btnSubmit"
                Content="Submit" Width="110" Height="34"
                HorizontalAlignment="Center" VerticalAlignment="Top"
                Click="btnSubmit_Click"/>
    </Grid>
</Window>
```

C# Code (MainWindow.xaml.cs)

```
using System;
using System.Windows;

namespace Lab05
{
    public partial class MainWindow : Window
    {
        public MainWindow()
        {
            InitializeComponent();

            // Rule (b): Male checked by default
            rbMale.IsChecked = true;

            // Rule (c): DOB range 1 Oct 1950 -> today, default = today
            dpDob.DisplayDateStart = new DateTime(1950, 10, 1);
            dpDob.DisplayDateEnd   = DateTime.Today;
            dpDob.SelectedDate     = DateTime.Today;
        }

        private void btnSubmit_Click(object sender, RoutedEventArgs e)
        {
            // Rule (a): mandatory fields
            if (string.IsNullOrWhiteSpace(txtFirstName.Text))
            {
                MessageBox.Show("First Name is mandatory!", "Validation Error",
                    MessageBoxButton.OK, MessageBoxImage.Warning);
                txtFirstName.Focus();
                return;
            }
            if (string.IsNullOrWhiteSpace(txtLastName.Text))
            {
                MessageBox.Show("Last Name is mandatory!", "Validation Error",
                    MessageBoxButton.OK, MessageBoxImage.Warning);
                txtLastName.Focus();
                return;
            }

            // Extra safety: typed DOB must stay inside the allowed range
            DateTime? dob = dpDob.SelectedDate;
            if (dob == null || dob < new DateTime(1950, 10, 1) || dob > DateTime.Today)
```

```
            {
                MessageBox.Show("DOB must be between 1 Oct 1950 and today.",
                    "Validation Error", MessageBoxButton.OK, MessageBoxImage.Warning);
                return;
            }

            // Form is valid -> information message box
            MessageBox.Show("welcome " + txtFirstName.Text.Trim() + " " +
                            txtLastName.Text.Trim(),
                "Information", MessageBoxButton.OK, MessageBoxImage.Information);
        }
    }
}
```

Output



Figure 2.1: Form on start-up (Male selected, DOB = today)

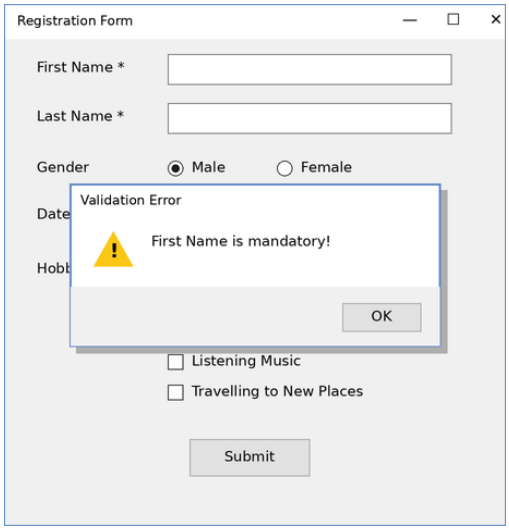


Figure 2.2: Validation message when mandatory fields are empty



Figure 2.3: Form filled with valid data

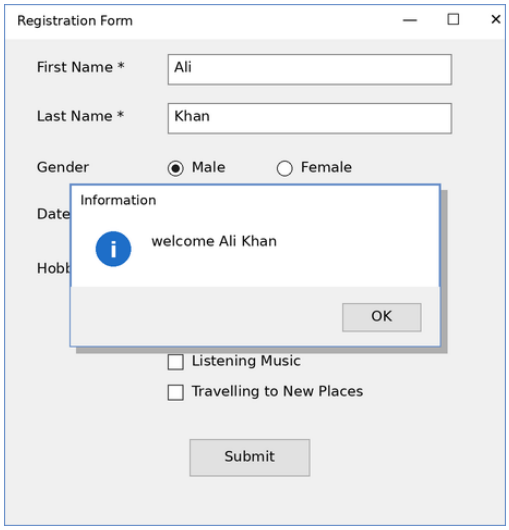


Figure 2.4: Information message box after clicking Submit

Conclusion

In this lab, a login page with hard-coded validation and a registration form with mandatory-field, default-value and date-range validation were created using WPF controls. The Click event was used to validate input and show a MessageBox with an information icon.